QUANTUM GATE PLAYGROUND

DAY 24 | 30-DAY QUANTUM COMPUTING CHALLENGE

Experiment objective

Apply X, Y, Z, H, S and T gates to a single qubit, compare output states, and visualize both measurement probabilities and phase-sensitive Bloch vectors.

1. Why this experiment matters

Some gates change a measured bit; others change only phase. A single histogram cannot show the full effect of a quantum gate. Comparing |0> and |+> as inputs reveals this difference.

2. Circuit and mathematics

Prepare |0>, optionally apply H to obtain $|+> = (|0> + |1>)/\sqrt{2}$, then apply one test gate and inspect the state. For $|\psi> = \alpha|0> + \beta|1>$, $P(0) = |\alpha|^2$ and $P(1) = |\beta|^2$.

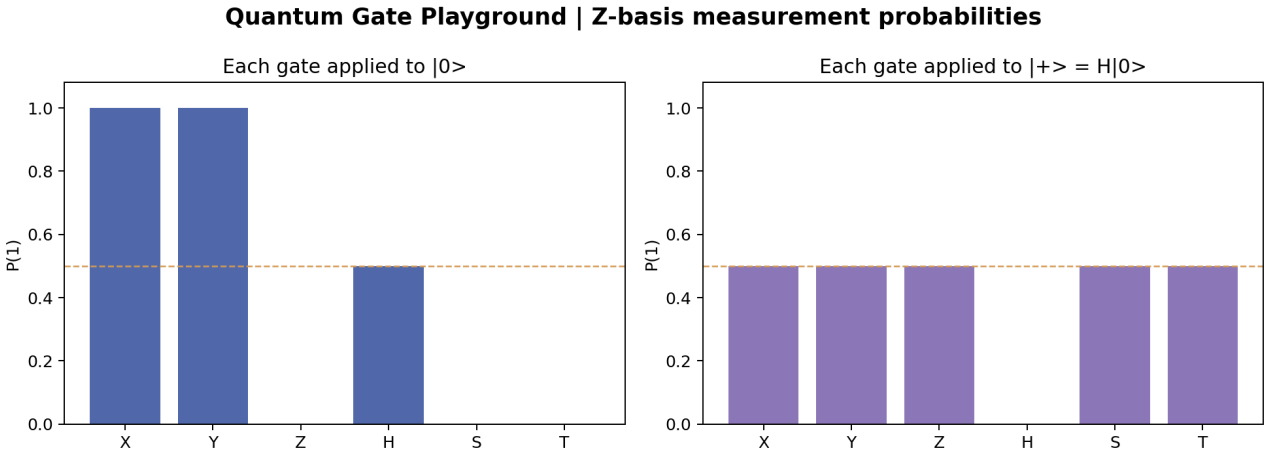
3. Gate reference

Gate	Action	From |0>
X	Bit flip	|1>
Y	Bit flip with phase	i|1>
Z	Sign flip on |1> amplitude	|0>
H	Basis transformation	$(\|0>+\|1>)/\sqrt{2}$
S	Phase $\pi/2$ on |1>	|0>
T	Phase $\pi/4$ on |1>	|0>

4. Phase is the key distinction

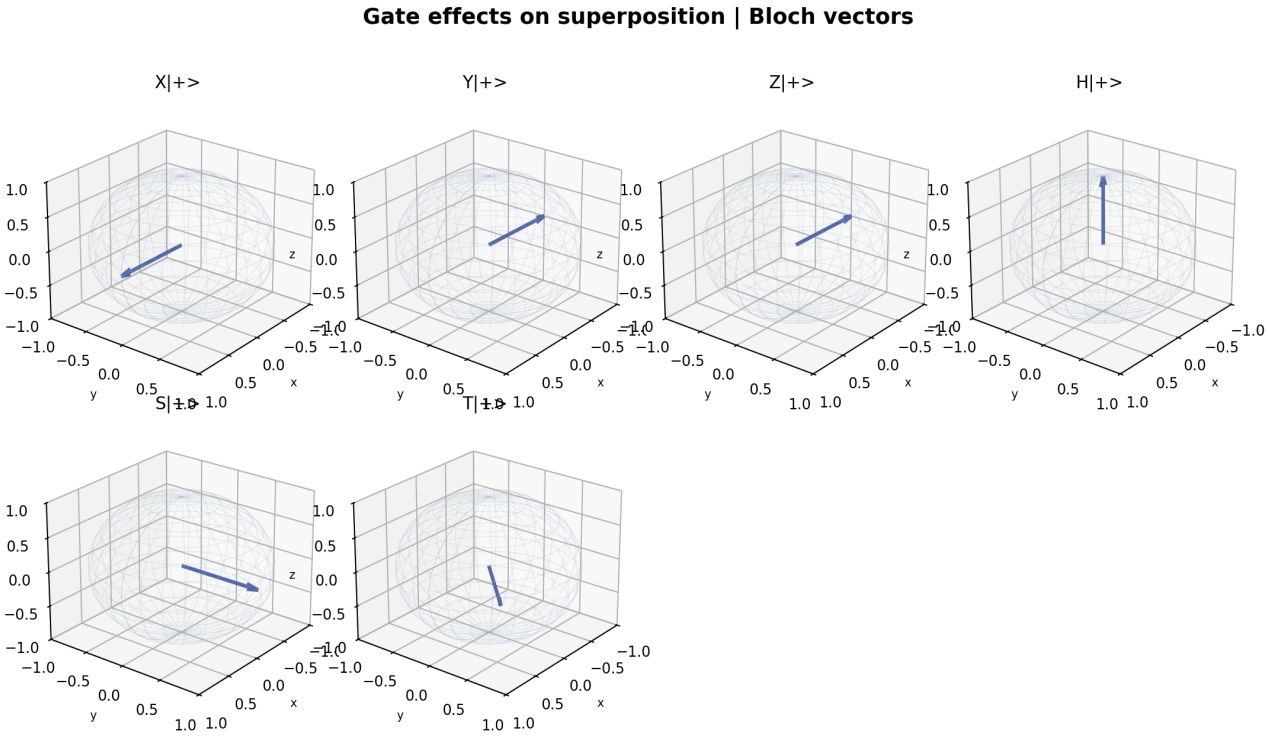
Z, S and T leave |0> unchanged, because the |1> amplitude is zero. Applied to |+>, Z creates |–>, S creates $(|0> + i|1>)/\sqrt{2}$, and T creates $(|0> + e^{(i\pi/4)}|1>)/\sqrt{2}$. All three still measure 0 or 1 with 50% probability in the Z basis.

5. Measurement probability experiment



Left: gates applied to |0>. Right: gates applied to |+>. S and T have no effect on Z-basis probabilities for either input, although they rotate the relative phase of |+>.

6. Bloch-sphere experiment



Bloch vectors distinguish states with identical Z-basis probabilities. In particular, S rotates |+> toward +y, T rotates it by 45° in the equatorial plane, and Z points it toward –x.

7. Run the experiment

• Install: **pip install numpy matplotlib notebook**.

• Open **quantum_gate_playground.ipynb** in Jupyter or VS Code.

• Execute gate definitions, statevector calculations and the two visualization sections.

• Change **sequence = "HST"** to test other combinations such as HH, HZ or HT.

• Compare output probabilities and Bloch vectors. Optional: install Qiskit for circuit verification.

8. Worked comparison

Experiment A: H|0> gives P(0) = P(1) = 0.5 and Bloch direction +x. **Experiment B:** ZH|0> gives the same probabilities but Bloch direction –x. Thus, the computational-basis histogram does not capture relative phase.

9. Common mistakes

Do not assume a gate is inactive because a measurement histogram stays unchanged. Do not confuse global phase with relative phase: Y|0> = i|1> is physically equivalent to |1>, but S|+> is not physically equivalent to |+>.

10. Limitations

The notebook uses ideal single-qubit statevectors and does not model measurement noise, gate errors or decoherence. A Bloch sphere represents a single-qubit state; entangled multi-qubit states require richer representations.

Summary: What to remember

X flips basis states; Y combines a flip and phase; Z changes the sign of the |1> component; H creates superposition from |0>; S and T rotate relative phase. Compare statevectors or Bloch vectors alongside probabilities to understand each gate.